

Q1 Linear Models Assignment

1.
$$SSE(b_0, b_1, b_2) = \sum_{i=1}^N (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2})^2$$

2. First order conditions

$$\frac{\partial SSE}{\partial b_0} = -2 \sum_i^N e_i = 0$$

$$\frac{\partial SSE}{\partial b_1} = -2 \sum_i^N e_i z_{i1} = 0$$

$$\frac{\partial SSE}{\partial b_2} = -2 \sum_i^N e_i z_{i2} = 0$$

3. From $\sum_{i=1}^N e_i = 0$, the average error is zero: $\bar{e} = 0$

From the other 2 FOCs, the residuals are uncorrelated to each regressor:

$$\sum_i^N e_i z_{i1} = 0 \quad \sum_i^N e_i z_{i2} = 0$$

4. Using $\sum_i^N e_i = 0$,

$$\begin{aligned} 0 &= \sum_i^N (y_i - b_0 - b_1 z_{i1} - b_2 z_{i2}) \\ &= \sum_i^N y_i - N b_0 - b_1 \sum_i^N z_{i1} - b_2 \sum_i^N z_{i2} \end{aligned}$$

Because z 's are centered, $\sum_i z_{ij} = 0$. Hence $b_0^* = \bar{y}$

5. After eliminating $b_0 = \bar{y}$ and centering y as $\tilde{y}_i = y_i - \bar{y}$:

$$\begin{bmatrix} \sum z_{i1}^2 & \sum z_{i1}z_{i2} \\ \sum z_{i1}z_{i2} & \sum z_{i2}^2 \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} \sum z_{i1}\tilde{y}_i \\ \sum z_{i2}\tilde{y}_i \end{bmatrix}$$

6.
$$\begin{bmatrix} \text{Var}(x_1) & \text{Cov}(x_1, x_2) \\ \text{Cov}(x_1, x_2) & \text{Var}(x_2) \end{bmatrix} \begin{bmatrix} b_1 \\ b_2 \end{bmatrix} = \begin{bmatrix} \text{Cov}(x_1, y) \\ \text{Cov}(x_2, y) \end{bmatrix}$$

Matrix A is the covariance matrix of the regressors; vector c is their covariance with y . OLS finds b so residuals are orthogonal to the regressors.